

Qatar-1b

R-0244

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180516 · created 2026-07-17T05:39:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458254.8001 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.012
Transit depth (Rp/Rs)^2	1.66 +/- 0.3 [%]
Orbital Inclination (inc)	85.41 +/- 1.4
Airmass coefficient 1 (a1)	1.287 +/- 0.014
Airmass coefficient 2 (a2)	-0.1316 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.41
Optimal Annulus	10.13
Transit Duration (day)	0.0722 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.129 $\pm$ 0.012 vs 0.1463 (deviation 1.28 σ)
Duration fit vs published	0.0722 d vs 0.0692 d (deviation 0.43 σ)
Mid-time O–C	3.2 min
Depth SNR	9.6
Residual scatter	1.09 %

Light curve

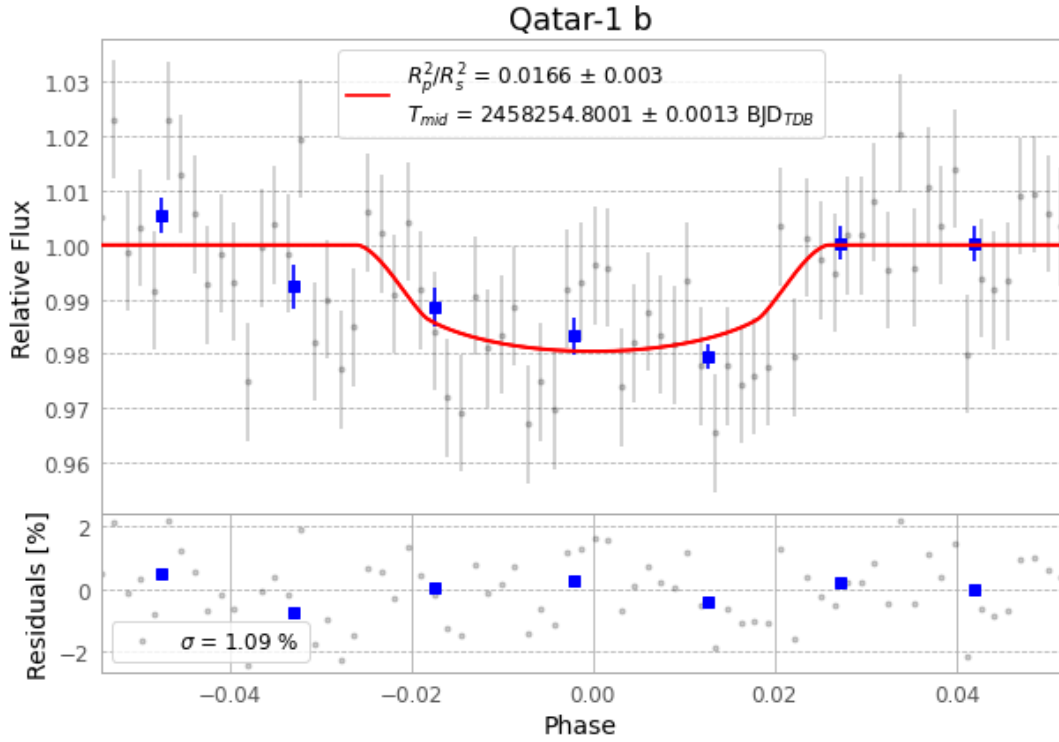


Figure 1 — FinalLightCurve_Qatar-1 b_2018-05-15.png (SHA-256 d787b1f27da5885c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 220], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 354b1ba7dffa0ef6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), Qatar-1180516052112.FITS → Qatar-1180516085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180516.log (f16a9aa9517f...), FinalLightCurve_Qatar-1 b_2018-05-15.png (d787b1f27da5...), FinalLightCurve_Qatar-1 b_2018-05-15.csv (3d113662ee8e...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:39Z.